



SURFACE REPRESENTATION

WHAT IS SURFACE REPRESENTATION

- Surface representation is a method used in computer graphics to describe the outer boundary of a 3D object.
- Instead of storing the entire volume, the system only stores the surface that defines the shape.

Example

A cube can be represented using 6 square surfaces instead of storing the full solid cube.

Key Idea

Only the outer surface is needed to render and visualize objects.

TYPES OF SURFACE REPRESENTATION

Main methods used to represent surfaces:

- Polygon Surface Representation
- Quadric Surface Representation
- Parametric Surface Representation
- Implicit Surface Representation

Each method represents surfaces using different mathematical approaches.

1. POLYGON SURFACE REPRESENTATION

Most commonly used method.

Definition

Objects are represented using polygons (usually triangles or quadrilaterals).

Example

A sphere can be approximated using many small triangles.

Advantages

- Easy to render
- Used in real-time graphics
- Supported by GPUs

Used in

- Games
- 3D modeling software
- VR applications

2. QUADRIC SURFACE REPRESENTATION

Quadric surfaces are defined using second-degree equations.

Examples

- Sphere
- Cylinder
- Cone
- Ellipsoid
- Paraboloid

General equation

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$$

Advantages

- Smooth mathematical surfaces
- Accurate representation

3. PARAMETRIC SURFACE REPRESENTATION

Surfaces defined using parameters (u, v).

General form:

- $S(u,v)=(x(u,v),y(u,v),z(u,v))$
 - $x = f(u,v)$
 - $y = g(u,v)$
 - $z = h(u,v)$

Examples

- Bézier surfaces
- B-spline surfaces
- NURBS surfaces

Used in

- Car body design
- Aircraft modeling
- Industrial CAD systems

4. IMPLICIT SURFACE REPRESENTATION

Surface defined by an **implicit equation**:

$$F(x, y, z) = 0$$

All points satisfying the equation form the surface.

Example

Sphere:

- $x^2 + y^2 + z^2 - r^2 = 0$

Advantages

- Good for representing smooth and complex shapes

COMPARISON OF METHODS

Method	Description	Usage
Polygon	Surface made of polygons	Games, real-time graphics
Quadric	Mathematical surfaces	Engineering models
Parametric	Uses parameters (u,v)	CAD design
Implicit	Equation based surfaces	Scientific modeling

APPLICATIONS OF SR

Surface representation is used in:

- 3D animation
- Game development
- Virtual reality
- Medical imaging
- Autonomous vehicle simulations
- Digital twins

ADVANTAGES OF SR

- Efficient object representation
- Faster rendering
- Easy to manipulate shapes
- Compatible with graphics hardware

CONCLUSION

- Surface representation is a fundamental concept in computer graphics that enables efficient modeling and visualization of 3D objects.
- Different techniques are used depending on the accuracy, complexity, and application requirements.
- Understanding these methods is essential for graphics programming, AI vision systems, and CAD applications.



THANK YOU